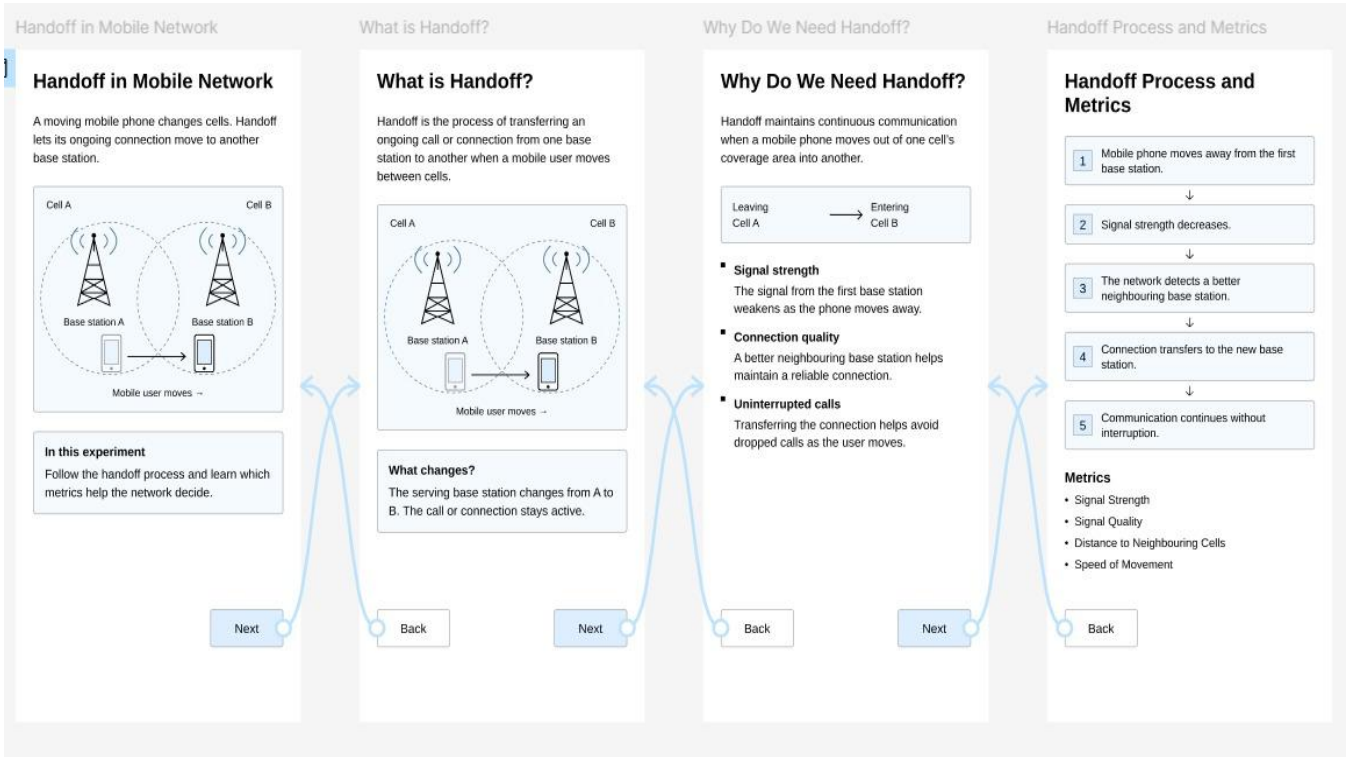
Exp 6:


Handoff in Mobile Network
Handoff in Mobile Network
A moving mobile phone changes cells. Handoff lets its ongoing connection move to another base station.
Cell A
Cell B
Base station A
Base station B
Mobile user moves →
In this experiment
Follow the handoff process and learn which metrics help the network decide.
Next
What is Handoff?
What is Handoff?
Handoff is the process of transferring an ongoing call or connection from one base station to another when a mobile user moves between cells.
Cell A
Cell B
Base station A
Base station B
Mobile user moves →
What changes?
The serving base station changes from A to B. The call or connection stays active.
Back
Next
Why Do We Need Handoff?
Why Do We Need Handoff?
Handoff maintains continuous communication when a mobile phone moves out of one cell's coverage area into another.
Leaving Cell A
Entering Cell B
Signal strength
The signal from the first base station weakens as the phone moves away.
Connection quality
A better neighbouring base station helps maintain a reliable connection.
Uninterrupted calls
Transferring the connection helps avoid dropped calls as the user moves.
Back
Next
Handoff Process and Metrics
Handoff Process and Metrics
1 Mobile phone moves away from the first base station.
2 Signal strength decreases.
3 The network detects a better neighbouring base station.
4 Connection transfers to the new base station.
5 Communication continues without interruption.
Metrics
Signal Strength
Signal Quality
Distance to Neighbouring Cells
Speed of Movement
Back